MSG No. 140.547 $I4/mc'm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8,13,14,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[17,18,19,20,25,26,27,28]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,22,23,24,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,9,10,22,23,31,32]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,11,12,21,24,29,30]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,8,13,14,18,19,27,28]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[6,7,15,16,17,20,25,26]`

Table 3: Wyckoff site: `4c`, site symmetry: `4/m..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[25,26,27,28,29,30,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `m.m'm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,8,27,28,31,32]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,6,7,25,26,29,30]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,13,14,18,19,22,23]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[11,12,15,16,17,20,21,24]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,11,21,31]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,10,22,30]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,9,23,29]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,12,24,32]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,15,17,27]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[6,14,18,26]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[7,13,19,25]`

8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,16,20,28]

Table 6: Wyckoff site: 8f, site symmetry: 4..

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,4]
2	$[0, 0, -z]$	[5,6,7,8]
3	$\left[0, 0, \frac{1}{2} - z\right]$	[9,10,11,12]
4	$\left[0, 0, z + \frac{1}{2}\right]$	[13,14,15,16]
5	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	[17,18,19,20]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	[21,22,23,24]
7	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[25,26,27,28]
8	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[29,30,31,32]

Table 7: Wyckoff site: 8g, site symmetry: 2.m'm'

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,4,31,32]
2	$\left[\frac{1}{2}, 0, z\right]$	[2,3,29,30]
3	$\left[0, \frac{1}{2}, -z\right]$	[5,8,27,28]
4	$\left[\frac{1}{2}, 0, -z\right]$	[6,7,25,26]
5	$\left[0, \frac{1}{2}, \frac{1}{2} - z\right]$	[9,10,22,23]
6	$\left[\frac{1}{2}, 0, \frac{1}{2} - z\right]$	[11,12,21,24]
7	$\left[0, \frac{1}{2}, z + \frac{1}{2}\right]$	[13,14,18,19]
8	$\left[\frac{1}{2}, 0, z + \frac{1}{2}\right]$	[15,16,17,20]

Table 8: Wyckoff site: 8h, site symmetry: m.2'm'

No.	position	mapping
1	$\left[x, x + \frac{1}{2}, 0\right]$	[1,8,27,32]
2	$\left[\frac{1}{2} - x, x, 0\right]$	[2,7,26,29]
3	$\left[x + \frac{1}{2}, -x, 0\right]$	[3,6,25,30]
4	$\left[-x, \frac{1}{2} - x, 0\right]$	[4,5,28,31]
5	$\left[x, \frac{1}{2} - x, \frac{1}{2}\right]$	[9,14,19,22]
6	$\left[-x, x + \frac{1}{2}, \frac{1}{2}\right]$	[10,13,18,23]
7	$\left[x + \frac{1}{2}, x, \frac{1}{2}\right]$	[11,16,17,24]
8	$\left[\frac{1}{2} - x, -x, \frac{1}{2}\right]$	[12,15,20,21]

Table 9: Wyckoff site: 16i, site symmetry: ..2'

No.	position	mapping
1	$\left[x, x, \frac{1}{4}\right]$	[1,11]
2	$\left[-x, x, \frac{1}{4}\right]$	[2,10]
3	$\left[x, -x, \frac{1}{4}\right]$	[3,9]

continued ...

Table 9

No.	position	mapping
4	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[4,12]`
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	`[5,15]`
6	$\left[x,\ -x,\ \frac{3}{4}\right]$	`[6,14]`
7	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[x,\ x,\ \frac{3}{4}\right]$	`[8,16]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[17,27]`
10	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[18,26]`
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[19,25]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[20,28]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[21,31]`
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[22,30]`
15	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[23,29]`
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[24,32]`

Table 10: Wyckoff site: `16j`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,9]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,11]`
3	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[3,12]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,10]`
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[5,13]`
6	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[6,15]`
7	$\left[0,\ x,\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[8,14]`
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[17,25]`
10	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[18,27]`
11	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[19,28]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[20,26]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[21,29]`
14	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[22,31]`
15	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[23,32]`
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[24,30]`

Table 11: Wyckoff site: `16k`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[-y,\ x,\ 0]$	`[2,7]`
3	$[y,\ -x,\ 0]$	`[3,6]`
4	$[-x,\ -y,\ 0]$	`[4,5]`
5	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[9,14]`
6	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[10,13]`
7	$\left[y,\ x,\ \frac{1}{2}\right]$	`[11,16]`

continued ...

Table 11

No.	position	mapping
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[12,15]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[17,24]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[18,23]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[19,22]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[20,21]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[25,30]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[26,29]
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[27,32]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ 0\right]$	[28,31]

Table 12: Wyckoff site: 16l, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	[1,32]
2	$\left[\frac{1}{2}-x,\ x,\ z\right]$	[2,29]
3	$\left[x+\frac{1}{2},\ -x,\ z\right]$	[3,30]
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	[4,31]
5	$\left[-x,\ \frac{1}{2}-x,\ -z\right]$	[5,28]
6	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	[6,25]
7	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	[7,26]
8	$\left[x,\ x+\frac{1}{2},\ -z\right]$	[8,27]
9	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[9,22]
10	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[10,23]
11	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[11,24]
12	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[12,21]
13	$\left[-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[13,18]
14	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[14,19]
15	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	[15,20]
16	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[16,17]

Table 13: Wyckoff site: 32m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[y,\ -x,\ -z]$	[6]
7	$[-y,\ x,\ -z]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[9]
10	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[10]
11	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[11]

continued ...

Table 13

No.	position	mapping
12	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[12]
13	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[21]
22	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[22]
23	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[23]
24	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[26]
27	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	[27]
28	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, -z\right]$	[28]
29	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[29]
30	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[30]
31	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[31]
32	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[32]